Probability exercises

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Easy

1. In a population where blood types are mutually exclusive categories, find the probability of a person being Type A OR Type B if the probability of Type A is 0.40 and the probability of Type B is 0.10.
2. In a study of 318 subjects, find the probability that a randomly selected person has an early age of onset (E) or no family history of mood disorders (A) if $P(E)=0.4434$, $P(A)=0.1981$, and the joint probability $P(E \cap A)=0.0881$.

Easy

3. Find the probability of a subject having an early onset (E) AND a negative family history (A) if the probability of early onset $P(E)=0.4434$ and the probability of a negative family history given early onset $P(A|E)=0.1986$.
4. In a class of 100 students where wearing glasses (E) and being a boy (B) are independent, find the probability of a student being a boy and wearing glasses if $P(B)=0.40$ and $P(E)=0.40$.

Easy

5. In a hospital with 1200 admissions where 750 are private (A), find the probability of a non-private admission if the probability of a private admission P(A)=0.625.
6. In an Alzheimer's screening, find the "sensitivity" (the probability of a positive test result T given the disease D) if the number of diseased subjects with a positive test is 436 and the total number of diseased subjects is 450.

Difficult

- Binomial formula: $P(X = x) = {}^{n}C_{x} p^{x} q^{n-x}$
- Data from the North Carolina State Center for Health Statistics show that 14 percent of mothers admit to smoking during pregnancy (p=0.14).
- In a random sample of 10 mothers (n=10), continue next slide…

Difficult

- what is the probability that exactly four of them admitted to smoking?
- what is the probability that more than four of them admitted to smoking?
- what is the probability that four or more of them admitted to smoking?
- what is the probability that less than four of them admitted to smoking?
- what is the probability that four or less of them admitted to smoking?

Difficult

- Combination values:
  - $^{10}C_0 = 120, {}^{10}C_1 = 10, {}^{10}C_2 = 45$, $^{10}C_3 = 120, {}^{10}C_4 = 210,$ $^{10}C_5 = 252$, $^{10}C_6 = 210$, …symmetric